9.1: Introduction to Probability

Definition.
A **sample space** is the set of all possible outcomes of a random process or experiment.
An **event** is a subset of a sample space.
If S is a finite sample space in which all outcomes are equally likely and E is an even in S, then the **probability of** E denoted $P(E)$, is

$$P(E) = \frac{\text{number of outcomes in } E}{\text{total number of outcomes in } S}$$

For any finite set A, $N(A)$ denotes the number of elements in A. Thus,

$$P(E) = \frac{N(E)}{N(S)}$$

Example. Consider a deck of 52 playing cards divided into four suits: ♠ ♥ ♣ ♦

What is the sample space?

S is the deck of 52 playing cards

What is the event that the chosen card is a red face card?

E = {J♥, Q♥, K♥, J♦, Q♦, K♦}

What is the probability that the chosen card is a red face card?

$$P(E) = \frac{N(E)}{N(S)} = \frac{6}{52} = \frac{3}{26} = 0.1\overline{153846} \approx 11.5\%$$

Example. Consider rolling a pair of 6-sided die.

Write the sample space S of possible outcomes.

$D_2 \backslash D_1$	1	2	3	4	5	6
1	11	21	31	41	51	61
2	12	22	32	42	52	62
3	13	23	33	43	53	63
4	14	24	34	44	54	64
5	15	25	35	45	55	65
6	16	26	36	46	56	66

Write the event E that the numbers appearing face up have a sum of 6 and find the probability of this event.

$$E = \{15, 24, 33, 42, 51\}$$

$$P(E) = \frac{N(E)}{N(S)} = \frac{5}{36} = 0.13\overline{8} \approx \boxed{13.\overline{8}\%}$$

The Number of Elements in a List
If m and n are integers and $m \leq n$, then there are $n - m + 1$ integers from m to n inclusive.

Example.

How many three-digit integers are divisible by 5?

100	105	110	115	...	990	995
$5 \cdot 20$	$5 \cdot 21$	$5 \cdot 22$	$5 \cdot 23$	...	$5 \cdot 198$	$5 \cdot 199$
$m = 20$						$n = 199$

$$n - m + 1 = 199 - 20 + 1 = \boxed{180}$$

What is the probability that $\underbrace{\text{a randomly chosen three-digit integer is divisible by 5?}}_{E}$

$$P(E) = \frac{N(E)}{N(S)} = \frac{180}{999 - 100 + 1} = \frac{180}{900} = \boxed{\frac{1}{5}}$$

Example. Let $A[1], A[2], \ldots, A[n]$ be a one-dimensional array, where n is a positive integer.

Suppose the array is cut at a middle value $A[m]$ so that two subarrays are formed:

$$A[1], A[2], \ldots, A[m] \text{ and } A[m+1], A[m+2], \ldots, A[n]$$

How many elements does each subarray have?

$m - 1 + 1 = m$ $\qquad$ $n - (m+1) + 1 = n - m$

What is the probability that a randomly chosen element of the array has an even subscript?

n even:

$$P(E) = \frac{N(E)}{N(S)} = \frac{n/2}{n} = \boxed{\frac{1}{2}}$$

n odd:

$$P(E) = \frac{N(E)}{N(S)} = \frac{\frac{n-1}{2}}{n} = \boxed{\frac{n-1}{2n}}$$

$$P(E) = \frac{\lfloor n/2 \rfloor}{n}$$ ← floor operator